Attention with Nonlinear Algebra

Explore replacing linear algebra in attention with alternative algebraic structures to probe expressive power boundaries.

Standard softmax attention is built entirely on linear algebra — dot products, matrix multiplications, and softmax normalization. This phase asks: what happens when you replace the core QK^T computation with polynomial kernels, Fourier transforms, or tropical (max-plus) algebra? The theoretical and empirical implications are largely unexplored.

CORE TASKS

- — Implement attention using polynomial kernels, Fourier transforms, or tropical algebra instead of standard QK^T scoring.
- — Test the effect on argmax-style broadcasting, numerical stability, and sequence length scaling behavior.
- — Analyze expressive power theoretically: which function classes can new attention approximate that standard cannot?
- — Build a benchmark suite across 5+ controlled synthetic sequence tasks.

MATHEMATICAL FORMULATION

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Standard: A = softmax(QK^T / sqrt(d)) · V
Polynomial: A = softmax(phi(Q)phi(K)^T) · V, phi(x) = [1, x, x^2, ..., x^p]
Fourier: A = softmax(Re[FFT(Q) · FFT(K)*] / sqrt(d)) · V
Tropical: A_ij = max_k(Q_ik + K_jk) - max_k(K_jk) [max-plus algebra]
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INSTRUMENTATION

- — Compare eigenvalue distributions and spectral norms across mechanisms.
- — Measure effective rank of attention weight matrices at each layer.
- — Track stability under heavy-tailed and adversarial input distributions.
- — Profile FLOPs vs. approximation error tradeoffs at sequence lengths 128–8192.

DELIVERABLES

Type	Description
Code	Unified attention API supporting all variants with shared evaluation harness

Theory	Formal proof or disproof: which variant(s) can express functions beyond softmax-linear attention
Analysis	Spectral analysis report with eigenvalue plots and rank collapse metrics
Writeup	Comparative study in paper format with ablation tables

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RESEARCH INSIGHT

Most attention research still uses vanilla linear algebra. Pushing into tropical or Fourier attention opens mechanistic questions most labs simply haven't had time to explore — and can yield genuine theoretical contributions.

Difficulty: Expert | Kernel methods · Tropical algebra · Spectral analysis · Expressive power